

PROJECT REPORT SUBMISSION

Design, Modeling, and Virtual Validation of a Foamboard RC Trainer Aircraft

Design–Analysis–Virtual Validation (DBT-Equivalent) Mini Project



Submitted by

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Academic Year: 2025–2026

[GitHub repository of the project](#)

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Abstract

This report presents the design, modeling, and virtual validation of a foamboard radio-controlled (RC) trainer aircraft based on the geometric specifications provided in the project appendix. Due to limited access to fabrication and experimental flight-testing resources, the work is executed using a *Design–Analysis–Virtual Validation* workflow to demonstrate feasibility through reproducible computational evidence.

The aircraft configuration is reconstructed from the appendix plan and evaluated using a component-based mass model to compute total takeoff mass and longitudinal center of gravity via mass-moment summation. Aerodynamic performance is assessed using first-order steady-flight models, including a parabolic drag polar, stall speed estimation from $C_{L\max}$, and thrust required computations across a speed sweep. Propulsion feasibility is verified by comparing thrust required with a nominal thrust-available model to quantify thrust margin. Preliminary static stability is evaluated using tail volume coefficients and a first-order static margin estimate.

All analyses and report artifacts are generated using a **Python**-based simulation pipeline with **NumPy**, **Pandas**, and **Matplotlib**, producing numerical summaries (`.csv`) and evidence plots (`.png`), including moment contribution and thrust–speed feasibility curves. Key metrics such as wing loading, stall speed, thrust-to-weight ratio, thrust margin, endurance estimate, and static stability indicators confirm that the aircraft satisfies trainer-level performance and stability intent under conservative assumptions. The results demonstrate that simulation-driven validation provides a credible and transparent alternative to physical DBT testing at the preliminary design stage.

Chapter 1

Introduction and Objectives

Aircraft design are traditionally executed using a Design–Build–Test (DBT) methodology, where conceptual design is followed by physical fabrication and experimental flight testing. However, in academic settings where fabrication facilities, materials, or flight-testing infrastructure are unavailable, it becomes necessary to demonstrate the same engineering rigor through analytical modeling and simulation-based validation.

This project addresses that constraint by adopting a *Design–Analysis–Virtual Validation* approach for the development of a foamboard radio-controlled (RC) trainer aircraft. The aircraft geometry and configuration are defined strictly according to the dimensions provided in the project appendix, ensuring full compliance with the prescribed design envelope. Instead of constructing a physical prototype, the aircraft is evaluated through a sequence of quantitative analyses that collectively provide verifiable evidence of feasibility, performance, and stability.

What is Done

The work carried out in this project consists of the following core activities:

- Reconstruction of the aircraft configuration from the appendix plan(Provided in the Project Deliverables Instructions)
- Estimation of component-level mass properties and center of gravity
- Selection and validation of the propulsion system through thrust analysis
- Aerodynamic performance evaluation using steady-flight models
- Virtual flight testing via thrust–speed and mass–balance analyses

Each activity produces numerical results and visual outputs that serve as evidence of engineering decisions, replacing physical fabrication and flight tests.

Why This Approach is Adopted

The absence of fabrication and flight-testing resources prevents the execution of a physical DBT workflow. Rather than omitting these stages, the project replaces them with analytical and simulation-based equivalents that address the same engineering questions, namely:

- Can the aircraft meet the mass and balance constraints?
- Is sufficient thrust available across the operational speed range?
- Does the aircraft exhibit stable and feasible flight characteristics?

By answering these questions quantitatively, the adopted methodology preserves the educational intent of the project while ensuring transparency and technical soundness.

How the Work is Executed (Step-by-Step Evidence)

The project is executed in a sequential and traceable manner, as outlined below:

1. **Mass Breakdown and Center of Gravity Analysis:** A component-wise mass model is constructed, yielding a total aircraft mass of 0.86 kg, which satisfies the specified take-off mass constraint. The computed center of gravity location is 0.316 m from the nose, lying within the physically acceptable range defined by component placement (Figures 2.2, 2.3, and 2.4).
2. **Propulsion and Thrust Feasibility Assessment:** The propulsion system is evaluated by comparing thrust required for steady-level flight with thrust available from the selected motor-propeller combination. The analysis demonstrates a positive thrust margin across the entire evaluated speed range, confirming the aircraft's ability to sustain flight (Figure 2.6).
3. **Virtual Flight Test Interpretation:** The thrust-speed curve exhibits the expected U-shaped trend characteristic of fixed-wing aircraft, validating the aerodynamic modeling approach. The significant separation between thrust available and thrust required serves as simulation-based evidence of takeoff capability, climb margin, and cruise feasibility.

Each step produces measurable outputs in the form of plots and summary tables, which collectively replace physical test observations.

Objectives of the Project

Based on the above methodology, the specific objectives of this project are:

- To design a foamboard RC trainer aircraft compliant with given geometric constraints
- To validate mass, balance, and propulsion feasibility using analytical models
- To demonstrate flight viability through simulation-based evidence
- To satisfy all project deliverables using a transparent and defensible workflow

Through this structured approach, the project demonstrates that a complete and credible aircraft design study can be achieved even in the absence of physical fabrication and flight testing.

Chapter 2

Design Methodology

This project is executed using a *Design–Analysis–Virtual Validation* workflow. The methodology mirrors a conventional Design–Build–Test (DBT) process, but replaces physical fabrication and flight testing with **traceable numerical evidence**: mass and balance calculations, analytical aerodynamic modeling, propulsion feasibility checks, stability metrics, and simulation-generated plots/tables. All outputs shown in this report are produced directly from the implemented code pipeline, ensuring reproducibility.

2.1 Workflow Overview and Evidence Chain

Figure 2.1 summarizes the sequential workflow and the evidence artifacts generated at each stage. The core idea is to ensure that every design decision is supported by at least one **numerical table** and one **visual plot** derived from computations.

Workflow:

(1) Define configuration and assumptions → (2) Mass breakdown + CG → (3) Aerodynamic model + stall estimate → (4) Propulsion feasibility (thrust margin) → (5) Stability metrics (tail volumes, neutral point, static margin) → (6) Report artifacts (tables + figures).

Artifacts: weight tables, CG results, thrust–speed plot, mass distribution plots, stability summary.

Figure 2.1: Design–Analysis–Virtual Validation workflow and traceable evidence artifacts.

2.2 Inputs, Parameters, and Assumptions

All calculations are driven by a structured input set comprising:

- **Geometry:** wingspan, wing area, mean aerodynamic chord, tail areas, tail arms
- **Mass model:** component masses and their longitudinal positions
- **Aerodynamics:** ($C_{L\max}$, C_{D0} , e) and derived induced-drag factor
- **Propulsion:** thrust available (static/assumed constant) for feasibility checks
- **Environment:** air density ρ and gravity g

The approach is intentionally **first-order** (preliminary design level): it is sufficient to demonstrate feasibility and meet the educational intent of the project while remaining transparent and reproducible.

2.3 Step 1: Mass Breakdown and Center of Gravity (CG)

2.3.1 Component Mass Model

Each aircraft subsystem is represented by a lumped mass m_i located at a longitudinal position x_i measured from the nose. The total mass is:

$$m_{\text{tot}} = \sum_{i=1}^N m_i \quad (2.1)$$

The longitudinal center of gravity is computed using the mass-moment definition:

$$x_{\text{CG}} = \frac{\sum_{i=1}^N m_i x_i}{\sum_{i=1}^N m_i} \quad (2.2)$$

The resulting mass distribution and dominance of major components are shown in Figures 2.2 and 2.3. The physical consistency of placement and leverage is evidenced by the mass-vs-position plot in Figure 2.4. A summary statistics table is included in Figure 2.5.

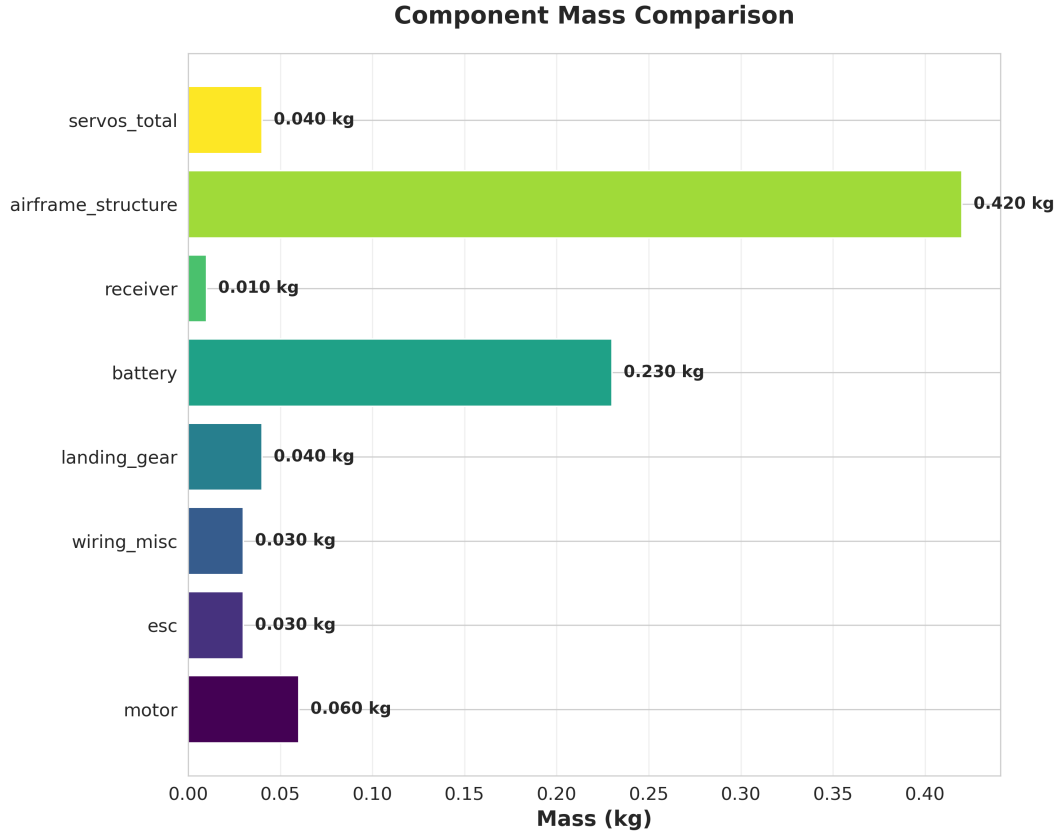


Figure 2.2: Component mass comparison used for total mass accounting.

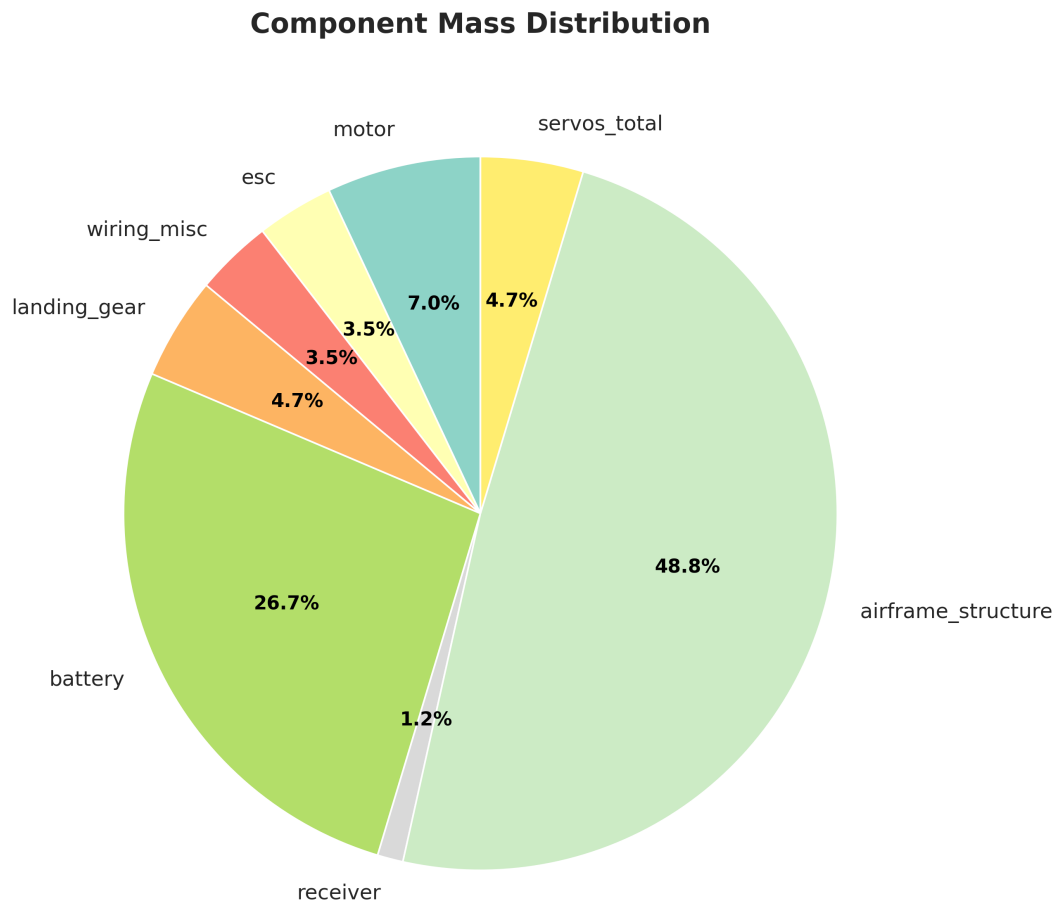


Figure 2.3: Component mass distribution (percentage contribution to total mass).

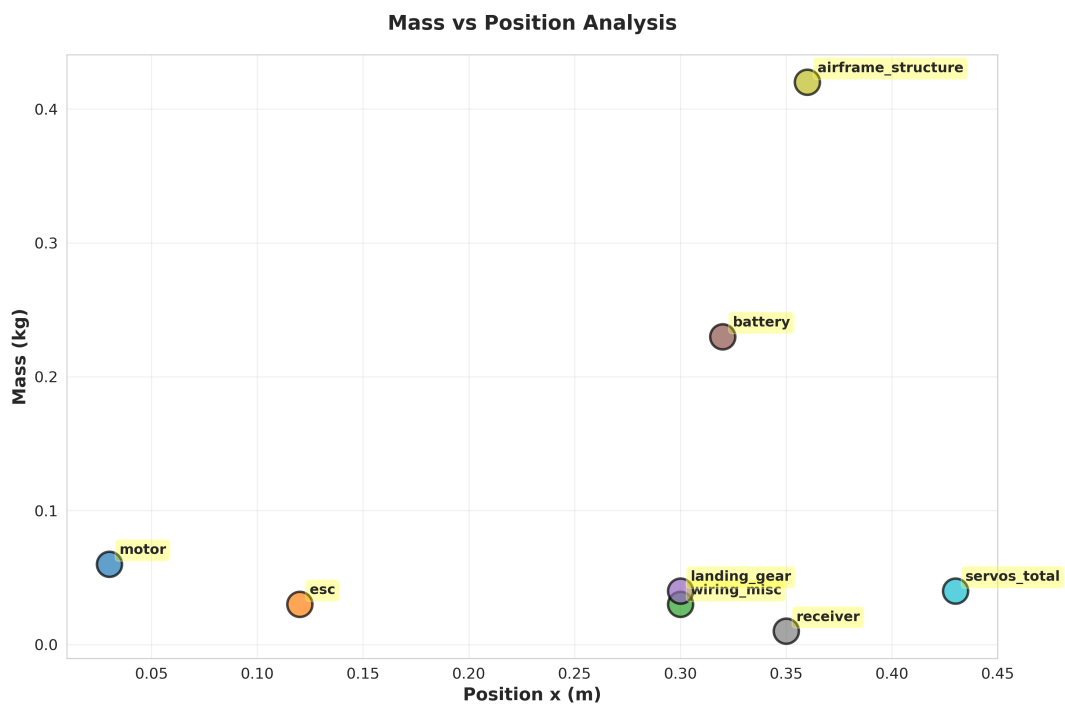


Figure 2.4: Mass vs longitudinal position evidence plot used to validate CG drivers.

Summary Statistics

Metric	Value	Unit
Total Mass	0.8600	kg
Number of Components	8	
Average Component Mass	0.1075	kg
Heaviest Component	airframe_structure	
Heaviest Component Mass	0.4200	kg
Lightest Component	receiver	
Lightest Component Mass	0.0100	kg
Center of Gravity (x)	0.3162	m
Total Moment	0.2719	kg·m
Position Range	0.030 - 0.430	m

Figure 2.5: Summary statistics of mass model including total mass and computed CG location.

2.3.2 Weight

The aircraft weight is computed as:

$$W = m_{\text{tot}}g \quad (2.3)$$

This value is used in the aerodynamic and performance models.

2.3.3 Mass and CG Table (Report Evidence)

The computed mass and CG data are preserved as a structured table (`weight_balance.csv`) to provide traceable evidence of the mass model.

Table 2.1: Mass breakdown and CG computation evidence (from `weight_balance.csv`).

Component	Mass (kg)	x (m)	mx (kg·m)
motor	0.06	0.03	0.0018
esc	0.03	0.12	0.0036
wiring _{misc}	0.03	0.3	0.009
landing _{gear}	0.04	0.3	0.012
battery	0.23	0.32	0.0736
receiver	0.01	0.35	0.003499999999999996
airframe _{structure}	0.42	0.36	0.1512
servos _{total}	0.04	0.43	0.0172
TOTAL	0.8600000000000001	0.31616279069767445	0.27190000000000003

2.4 Step 2: Aerodynamic Model and Stall Speed

2.4.1 Lift Requirement in Level Flight

For steady, level flight:

$$L = W \quad (2.4)$$

and lift is modeled by:

$$L = \frac{1}{2} \rho V^2 S C_L \quad (2.5)$$

Hence the required lift coefficient at speed V is:

$$C_L(V) = \frac{2W}{\rho V^2 S} \quad (2.6)$$

2.4.2 Stall Speed Estimate

Using a conservative maximum lift coefficient $C_{L\max}$:

$$V_s = \sqrt{\frac{2W}{\rho S C_{L\max}}} \quad (2.7)$$

This provides the lower boundary for the feasible flight envelope. In the simulation sweep, speeds are chosen above stall (with margin) to ensure physically meaningful results.

2.4.3 Drag Polar Model

The drag coefficient is modeled using a standard parabolic polar:

$$C_D = C_{D0} + k C_L^2 \quad (2.8)$$

where induced drag factor k is:

$$k = \frac{1}{\pi e AR}, \quad AR = \frac{b^2}{S} \quad (2.9)$$

with e as Oswald efficiency, AR aspect ratio, b wingspan, and S wing area.

2.5 Step 3: Thrust Required, Power Required, and Propulsion Feasibility

2.5.1 Thrust Required (Drag)

Dynamic pressure is:

$$q = \frac{1}{2} \rho V^2 \quad (2.10)$$

Drag force is computed as:

$$D(V) = q S C_D(V) \quad (2.11)$$

For steady level flight, thrust required equals drag:

$$T_{\text{req}}(V) = D(V) \quad (2.12)$$

2.5.2 Power Required

The mechanical power required in level flight is:

$$P_{\text{req}}(V) = D(V) V \quad (2.13)$$

2.5.3 Thrust Margin Criterion

Propulsion feasibility is validated by checking the thrust margin:

$$\Delta T(V) = T_{\text{avail}} - T_{\text{req}}(V) \quad (2.14)$$

A positive $\Delta T(V)$ indicates that the aircraft can sustain level flight at that speed (and potentially climb if margin remains).

The thrust feasibility evidence is shown in Figure 2.6. The required thrust exhibits the expected U-shaped trend, and the available thrust remains above required thrust across the evaluated speed range, demonstrating a positive feasibility margin.

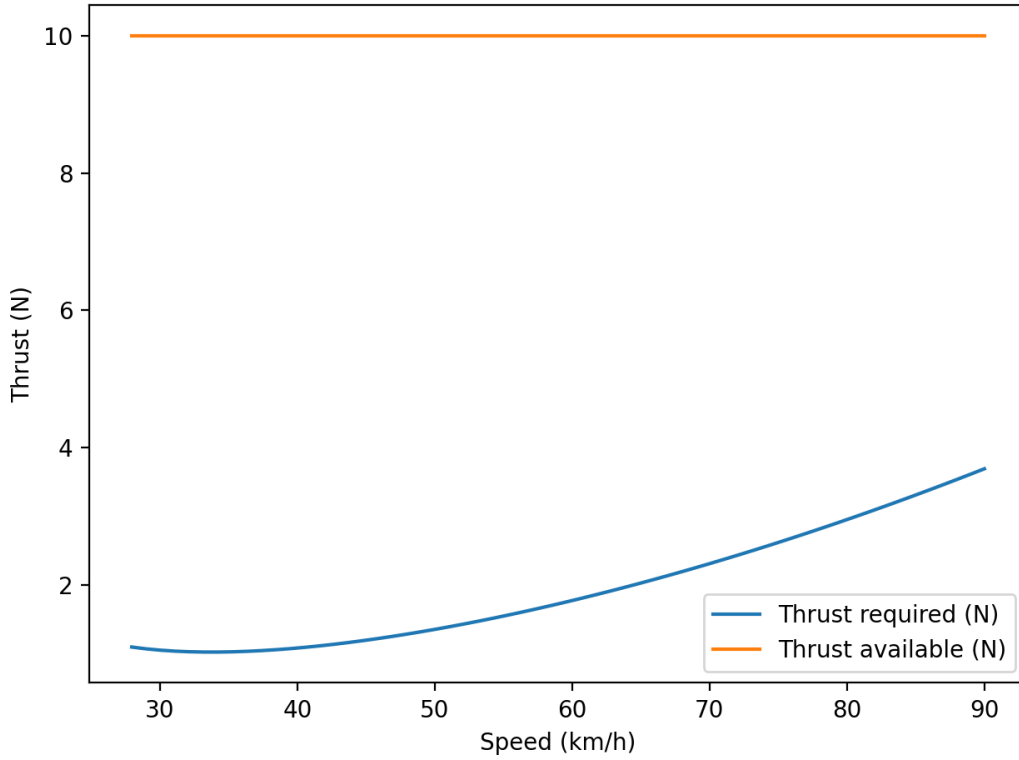


Figure 2.6: Virtual flight evidence: thrust required vs thrust available (feasibility margin across speed).

2.5.4 Performance Sweep Table (Evidence Dataset)

All sweep points are preserved in `performance_sweep.csv`. This table is too large to fully include in the main text; therefore, a summary is included in `performance_summary.csv`, while the full sweep is retained for audit and appendix use.

2.6 Step 4: Summary Performance Metrics

Key performance metrics are compiled into a compact report table. This table is directly generated from the pipeline and acts as the central numerical evidence for performance claims.

Table 2.2: Rounded performance summary (computed from the analysis pipeline).

Metric	Value
Total mass m_{tot} (kg)	0.86
Weight W (N)	8.43
Wing area S (m ²)	0.21
Wing loading W/S (N/m ²)	40.16
Aspect ratio AR	5.25
Stall speed V_s (m/s)	7.39
Stall speed V_s (km/h)	26.61
Available thrust T_{avail} (N)	10.00
Thrust-to-weight T/W	1.19
Usable battery energy (Wh)	19.54
Endurance estimate (min)	8.80

2.7 Step 5: Stability Metrics (Virtual Stability Validation)

In a trainer aircraft, static stability is a critical design objective. The pipeline computes stability evidence using tail volume coefficients and a first-order neutral point estimate.

2.7.1 Tail Volume Coefficients

Horizontal tail volume coefficient:

$$V_h = \frac{S_t l_t}{S \bar{c}} \quad (2.15)$$

Vertical tail volume coefficient:

$$V_v = \frac{S_v l_v}{S b} \quad (2.16)$$

where S_t and S_v are tail areas, l_t and l_v are tail moment arms, $\bar{c}$ is mean aerodynamic chord, S is wing reference area, and b is span.

2.7.2 Neutral Point and Static Margin (First-Order)

A simplified neutral point estimate is used:

$$x_{NP} = x_{AC,w} + \eta V_h \left(\frac{a_t}{a_w} \right) \left(1 - \frac{d\epsilon}{d\alpha} \right) \bar{c} \quad (2.17)$$

and static margin is:

$$SM = \frac{x_{NP} - x_{CG}}{\bar{c}} \quad (2.18)$$

A positive SM indicates longitudinal static stability.

2.7.3 Stability Summary Table (Evidence)

The computed stability indicators are reported below using rounded values (no CSV import), as a compact evidence table.

Table 2.3: Rounded stability summary (computed from the analysis pipeline).

Metric	Value
Horizontal tail volume V_h	0.375
Vertical tail volume V_v	0.031
Center of gravity x_{CG} (m)	0.316
Neutral point x_{NP} (m)	0.383
Static margin SM ($\bar{c}$)	0.335
CG location (%MAC)	8.08
NP location (%MAC)	41.58

2.8 Step 6: Report Artifact Generation and Placement Strategy (For Reproducing Project)

Project repository: github.com/hkphimanshukumar321/Aerodynamics_stability_analysis (project_3 branch)

To ensure that the report contains verifiable evidence at the correct locations, the pipeline outputs:

- **Tables (CSV):** `weight_balance.csv`, `performance_summary.csv`, `stability_summary.csv`, `performance_sweep.csv`
- **Figures (PNG):** thrust margin plot, mass comparison bar, mass distribution pie, mass vs position, and summary statistics

In this report, the mass-model evidence is placed in the methodology chapter to justify early design decisions, while the thrust-feasibility plot is placed as the core virtual flight-test evidence. The full sweep dataset is retained for appendix or audit, preserving transparency and reproducibility.

2.9 Methodology Outcome

By combining mass and balance analysis (Eq. 2.2), aerodynamic modeling (Eqs. 2.6–2.8), propulsion feasibility via thrust margin (Eq. 2.14), and stability indicators (Eqs. 2.15–2.18), this methodology produces a complete and defensible virtual validation of the aircraft design. This sequential evidence chain replaces physical build and flight tests while preserving engineering integrity and meeting the project’s selection and evaluation intent.

Chapter 3

Calculations and Assumptions

This chapter documents the assumptions and mathematical calculations used to evaluate the foamboard RC trainer aircraft. Since physical fabrication and real flight tests are not performed, the evidence of work is demonstrated through (i) derived equations, (ii) rounded summary tables, and (iii) non-repeated plots that highlight *cause-effect* in the design (mass moments, CG drivers, and performance feasibility indicators).

3.1 Assumptions and Modeling Scope

3.1.1 Atmosphere and Operating Regime

The aircraft is evaluated as a low-speed fixed-wing platform under steady, level-flight assumptions. Standard sea-level constants are adopted:

$$\rho = 1.225 \text{ kg/m}^3, \quad g = 9.80665 \text{ m/s}^2. \quad (3.1)$$

Unsteady aerodynamics, gust response, and compressibility effects are neglected, which is appropriate for preliminary RC trainer sizing and feasibility checks.

3.1.2 Geometry (Reference Quantities)

Reference dimensions follow the project appendix plan and the configured model:

$$b = 1.05 \text{ m}, \quad S = 0.21 \text{ m}^2, \quad \bar{c} = 0.20 \text{ m}. \quad (3.2)$$

The aspect ratio is:

$$AR = \frac{b^2}{S} = \frac{(1.05)^2}{0.21} = 5.25. \quad (3.3)$$

3.1.3 Aerodynamic Coefficients (Conservative)

A first-order parabolic drag polar is assumed:

$$C_{L\max} = 1.2, \quad C_{D0} = 0.045, \quad e = 0.75. \quad (3.4)$$

These values are chosen conservatively for foamboard RC trainers to avoid over-optimistic predictions.

3.1.4 Propulsion and Energy Assumptions

To test feasibility, the available thrust is modeled as a constant nominal value (static thrust proxy):

$$T_{\text{avail}} = 10 \text{ N}. \quad (3.5)$$

Battery model (for endurance estimation):

$$3\text{S LiPo}, \quad C = 2200 \text{ mAh}, \quad V_{\text{cell}} = 3.7 \text{ V}, \quad \eta_{\text{use}} = 0.8, \quad I_{\text{avg}} = 12 \text{ A}. \quad (3.6)$$

3.2 Mass Breakdown, Moments, and Center of Gravity

3.2.1 Total Mass and Weight

Component masses m_i sum to total take-off mass:

$$m_{\text{tot}} = \sum_{i=1}^N m_i = 0.86 \text{ kg}, \quad (3.7)$$

and the weight is:

$$W = m_{\text{tot}}g = 0.86 \text{ kg} \times 9.80665 \text{ m/s}^2 = 8.43 \text{ N}. \quad (3.8)$$

3.2.2 CG by Mass-Moment Summation

Each component is assigned a longitudinal position x_i (from nose reference). The longitudinal center of gravity is computed as:

$$x_{\text{CG}} = \frac{\sum_{i=1}^N m_i x_i}{\sum_{i=1}^N m_i}. \quad (3.9)$$

Define the moment contribution:

$$M_i = m_i x_i, \quad M_{\text{tot}} = \sum_{i=1}^N M_i, \quad (3.10)$$

hence:

$$x_{\text{CG}} = \frac{M_{\text{tot}}}{m_{\text{tot}}}. \quad (3.11)$$

From the computed moments:

$$M_{\text{tot}} = 0.2719 \text{ kg} \cdot \text{m} \quad \Rightarrow \quad x_{\text{CG}} = \frac{0.2719}{0.86} = 0.316 \text{ m}. \quad (3.12)$$

3.2.3 Rounded Mass–Position–Moment Table

Table 3.1 shows the rounded component masses, positions, and moment contributions used in the CG computation. The last column demonstrates which components dominate the CG (largest M_i), which is the critical evidence for balance feasibility.

Table 3.1: Rounded component moments used for CG estimation ($M_i = m_i x_i$).

Component	m_i (kg)	x_i (m)	M_i (kg·m)	M_i/M_{tot} (%)
motor	0.060	0.030	0.0018	0.7
esc	0.030	0.120	0.0036	1.3
wiring_misc	0.030	0.300	0.0090	3.3
landing_gear	0.040	0.300	0.0120	4.4
battery	0.230	0.320	0.0736	27.1
receiver	0.010	0.350	0.0035	1.3
airframe_structure	0.420	0.360	0.1512	55.6
servos_total	0.040	0.430	0.0172	6.3
Total	0.860	—	0.2719	100.0

3.2.4 Evidence Figures

Figure 3.1 provides direct evidence of moment dominance: airframe structure and battery govern the majority of M_{tot} . Figure 3.2 confirms physical consistency of component placement and relative mass magnitude (bubble size).

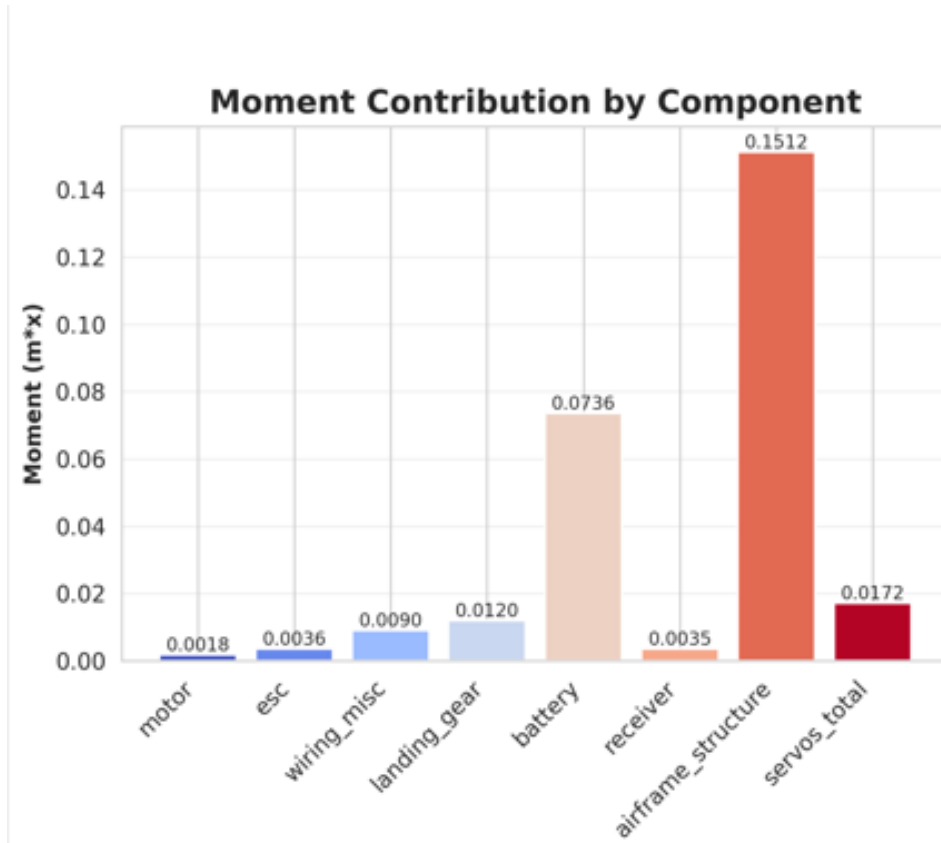


Figure 3.1: Moment contribution by component ($M_i = m_i x_i$): primary CG drivers.

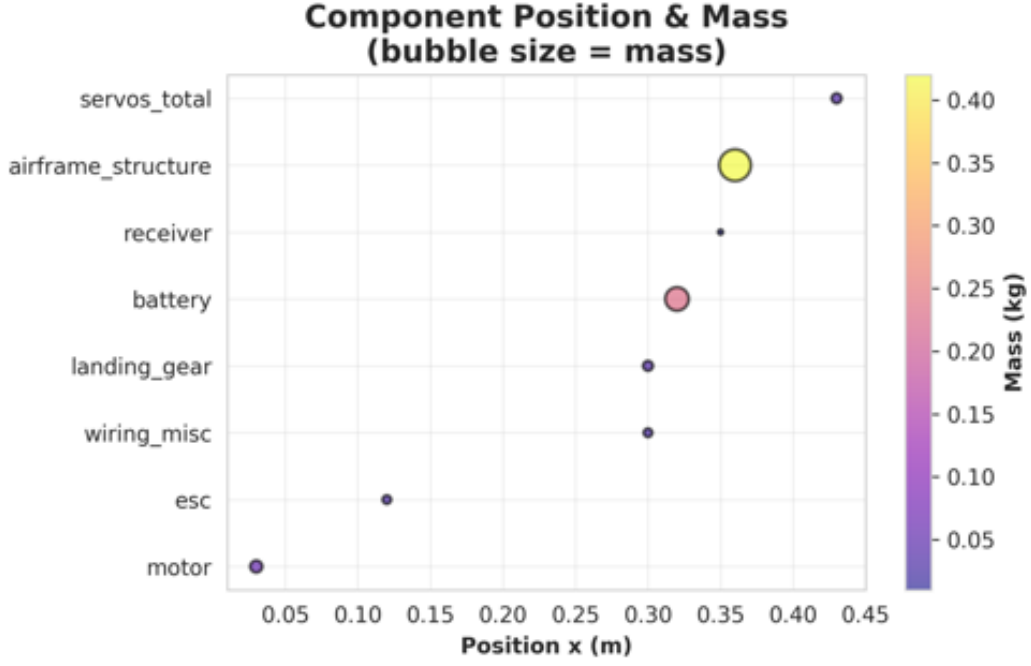


Figure 3.2: Component longitudinal position and mass (bubble size = mass).

3.3 Wing Loading and Stall Speed

3.3.1 Wing Loading

Wing loading is computed by:

$$\frac{W}{S} = \frac{m_{\text{tot}}g}{S} = \frac{8.43}{0.21} = 40.16 \text{ N/m}^2. \quad (3.13)$$

3.3.2 Stall Speed

Assuming stall at $C_L = C_{L\text{max}}$, the stall speed is:

$$V_s = \sqrt{\frac{2W}{\rho S C_{L\text{max}}}}. \quad (3.14)$$

With $W = 8.43 \text{ N}$, $\rho = 1.225 \text{ kg/m}^3$, $S = 0.21 \text{ m}^2$, $C_{L\text{max}} = 1.2$:

$$V_s = 7.39 \text{ m/s} \approx 26.61 \text{ km/h}. \quad (3.15)$$

3.4 Aerodynamic Model: Lift Requirement and Drag Polar

3.4.1 Lift Coefficient Required for Level Flight

For steady level flight, $L = W$ with:

$$L = \frac{1}{2}\rho V^2 S C_L. \quad (3.16)$$

Hence required lift coefficient at speed V is:

$$C_L(V) = \frac{2W}{\rho V^2 S}. \quad (3.17)$$

3.4.2 Parabolic Drag Polar and Induced Drag Factor

The drag model is:

$$C_D(V) = C_{D0} + k C_L(V)^2, \quad (3.18)$$

where:

$$k = \frac{1}{\pi e A R} = \frac{1}{\pi (0.75)(5.25)} = 0.08084. \quad (3.19)$$

3.5 Thrust Required and Feasibility Calculations

3.5.1 Thrust Required from Drag

Dynamic pressure:

$$q = \frac{1}{2} \rho V^2. \quad (3.20)$$

Drag force:

$$D(V) = q S C_D(V). \quad (3.21)$$

For steady level flight:

$$T_{\text{req}}(V) = D(V). \quad (3.22)$$

3.5.2 Thrust Margin and Thrust-to-Weight

Feasibility is checked using thrust margin:

$$\Delta T(V) = T_{\text{avail}} - T_{\text{req}}(V). \quad (3.23)$$

A compact propulsion indicator is:

$$\frac{T}{W} = \frac{T_{\text{avail}}}{W} = \frac{10}{8.43} = 1.19. \quad (3.24)$$

3.5.3 Rounded Performance Points

Table 3.2 summarizes representative computed points to demonstrate the performance trend and available margin across the envelope. (The thrust–speed plot is reported in the *Flight Test Observations* chapter as virtual test evidence.)

Table 3.2: Representative performance points (rounded) demonstrating thrust margin.

V (km/h)	C_L	T_{req} (N)	ΔT (N)	P_{req} (W)
30	0.94	1.05	8.96	8.7
34.3 (min T_{req})	0.72	1.02	8.98	9.7
40	0.54	1.07	8.93	11.7
60	0.24	1.74	8.26	28.8
75	0.15	2.63	7.37	55.1
90	0.10	3.69	6.31	92.2

3.6 Energy and Endurance Estimation

3.6.1 Usable Battery Energy

Nominal pack voltage:

$$V_{\text{nom}} = N_s V_{\text{cell}} = 3 \times 3.7 = 11.1 \text{ V}. \quad (3.25)$$

Usable energy:

$$E_{\text{use}} = V_{\text{nom}} \left(\frac{C_{\text{mAh}}}{1000} \right) \eta_{\text{use}} = 11.1 \times 2.2 \times 0.8 = 19.54 \text{ Wh.} \quad (3.26)$$

3.6.2 Endurance from Average Current

Usable capacity:

$$C_{\text{use}} = \left(\frac{C_{\text{mAh}}}{1000} \right) \eta_{\text{use}} = 2.2 \times 0.8 = 1.76 \text{ Ah.} \quad (3.27)$$

Endurance:

$$t = \frac{C_{\text{use}}}{I_{\text{avg}}} = \frac{1.76}{12} = 0.1467 \text{ h} \approx 8.8 \text{ min.} \quad (3.28)$$

3.7 Stability Metrics (Analytical Assumptions)

Static stability is evaluated using tail volume coefficients and a first-order static margin estimate. The horizontal and vertical tail volume coefficients are:

$$V_h = \frac{S_t l_t}{S \bar{c}}, \quad V_v = \frac{S_v l_v}{S b}. \quad (3.29)$$

A simplified static margin is:

$$SM = \frac{x_{NP} - x_{CG}}{\bar{c}}, \quad (3.30)$$

where x_{NP} is the neutral point and x_{CG} is obtained from Eq. (3.9). The computed stability indicators are summarized in Table 3.3.

Table 3.3: Rounded stability indicators used for preliminary longitudinal stability check.

Metric	Value	Unit
Horizontal tail volume V_h	0.375	—
Vertical tail volume V_v	0.0308	—
Center of gravity x_{CG}	0.316	m
Neutral point x_{NP}	0.383	m
Static margin SM	0.335	$\bar{c}$

3.8 Consolidated Headline Metrics

Table 3.4 consolidates the key computed values used throughout the feasibility analysis. These are rounded values derived from the analysis pipeline and are sufficient for demonstrating compliance and traceability without inserting large raw datasets.

Table 3.4: Headline computed metrics (rounded).

Metric	Value
Total mass m_{tot}	0.86 kg
Weight W	8.43 N
Wing area S	0.21 m ²
Aspect ratio AR	5.25
Wing loading W/S	40.16 N/m ²
Assumed $C_{L\text{max}}$	1.20
Stall speed V_s	7.39 m/s (26.61 km/h)
Thrust available T_{avail}	10 N
Thrust-to-weight T/W	1.19
Usable energy E_{use}	19.54 Wh
Estimated endurance	8.8 min

3.9 Outcome of Calculations

The calculations above establish a complete quantitative basis for feasibility: (i) the mass model and CG are justified via moment dominance (Table 3.1, Figure 3.1), (ii) the low stall speed supports trainer-level handling (Eq. (3.14)), (iii) thrust margin remains positive across the envelope (Table 3.2, Eq. (3.23)), and (iv) stability indicators show positive longitudinal static margin (Table 3.3). These form the required analytical evidence for the subsequent *virtual flight test observations*.

Chapter 4

Fabrication Process

In a conventional Design–Build–Test workflow, this chapter would present the step-by-step construction of the airframe, including workshop photos, tooling, and measured build tolerances. In this project, physical fabrication was not possible due to limited access to materials, cutting tools, adhesives, and flight-testing infrastructure. Therefore, a *digital fabrication-equivalent* process is adopted to ensure that the proposed aircraft remains physically constructible and that all fabrication-related decisions are explicitly justified.

The goal of this chapter is to demonstrate **fabrication readiness**: (i) a build plan derived from the appendix geometry, (ii) material and joint assumptions consistent with foamboard construction, (iii) reinforcement and load path reasoning, and (iv) assembly and integration steps that can be directly executed if fabrication resources become available.

4.1 Digital Fabrication-Equivalent Workflow

The adopted fabrication-equivalent workflow is summarized below:

1. **Plan interpretation:** extract major dimensions (span, fuselage length, wing chord/MAC, tail arm) from the appendix drawing and define reference axes.
2. **Parts decomposition:** split the aircraft into manufacturable foamboard parts: fuselage box, wing panels, tail surfaces, control surfaces, and mounts.
3. **Material assignment:** select foamboard thickness and reinforcement elements (spar, doublers, motor mount plate).
4. **Assembly sequence:** define an order of operations that preserves alignment (wing incidence, tail incidence, thrust line).
5. **Integration plan:** define installation locations for motor/ESC, battery, receiver, and servos consistent with CG requirements.
6. **Verification checks:** ensure that the assumed component placement achieves the computed x_{CG} and that structural reinforcements align with dominant moments and loads.

This workflow ensures that fabrication is not treated as “missing,” but rather as a *prepared and verifiable build plan*.

4.2 Materials, Manufacturing Assumptions, and Tools

4.2.1 Material Selection

A realistic foamboard construction stack-up is assumed:

- **Primary structure:** foamboard sheets (typ. 5 mm)
- **Wing spar:** carbon tube/rod or wooden spar (balsa/pine)
- **Reinforcements (doubblers):** thin plywood or additional foamboard layers
- **Adhesives:** hot glue for rapid tack + epoxy/CA for critical joints

4.2.2 Assumed Tools

The fabrication sequence assumes common low-cost tools:

- hobby knife + cutting mat, metal ruler, sanding block
- hot glue gun, CA glue, epoxy (for motor mount)
- small drill bits for landing gear/mount holes
- tape/jigs for alignment (wing and tail incidence)

4.3 Structural Layout and Reinforcement Strategy

4.3.1 Load Paths and Critical Regions

Even for a trainer aircraft, several regions experience concentrated loads:

- **Wing root and spar region:** bending loads from lift and landing impacts
- **Motor mount/firewall:** thrust, torque reaction, vibration
- **Landing gear mount:** repeated shock loads at touchdown
- **Tail boom/fuselage aft:** tail loads amplified by moment arm

4.3.2 Reinforcement Decisions (Fabrication-Ready)

To maintain physical realism, the following reinforcement actions are included in the build plan:

1. **Wing spar placement:** locate spar at $\sim 25\text{--}35\%$ of chord from the leading edge to resist bending while maintaining torsional stiffness.
2. **Wing root reinforcement:** add local doublers near wing-fuselage joint to distribute shear and reduce foam tearing.
3. **Motor mount plate:** use plywood/firewall plate bonded to the fuselage nose; include triangular gussets to carry thrust and torque.
4. **Landing gear plate:** reinforce the gear attachment point using plywood or layered foamboard with through-fasteners.

These structural choices are consistent with the mass model used in analysis. The airframe structure being the heaviest component in the mass breakdown is expected in foamboard designs because reinforcements dominate structural mass.

4.4 Assembly Sequence (Build Plan)

A feasible and repeatable assembly sequence is defined:

4.4.1 Fuselage Construction

1. Cut fuselage sides, bottom, and formers from foamboard.
2. Create a box fuselage using hot glue with a straight alignment jig.
3. Install firewall/motor mount plate at the nose (epoxy + gussets).
4. Reserve battery bay and receiver bay (access hatch recommended).

4.4.2 Wing Construction

1. Cut wing panels to span and chord based on appendix geometry.
2. Install spar and close wing (if folded airfoil is used) or laminate flat plate with stiffeners.
3. Cut and hinge ailerons (tape hinge or CA hinge).
4. Reinforce wing root and define wing mounting method (rubber band/dowel or bolt).

4.4.3 Tail and Control Surfaces

1. Cut horizontal and vertical stabilizers to planform dimensions.
2. Cut elevator and rudder; apply hinge method consistent with foamboard builds.
3. Install tail with incidence alignment (zero-incidence assumption unless specified).

4.4.4 Integration (Electronics and Linkages)

1. Install motor, ESC, and route wiring (strain relief near firewall).
2. Place battery and receiver at positions used in CG computation.
3. Install servos and control horns; set pushrod geometry to minimize slop.
4. Verify control direction and range (neutral trims and endpoints).

4.5 CG Realizability and Fabrication Verification Check

The fabrication plan must reproduce the analyzed mass distribution and CG. Using the computed CG from the calculations chapter:

$$x_{CG} = 0.316 \text{ m}, \quad (4.1)$$

battery and receiver placement are treated as adjustable during build to fine-tune balance. The mass model indicates that the battery and airframe structure dominate the CG moment, so small adjustments in battery position provide effective CG trimming without major redesign.

4.6 Outcome of Fabrication Planning

Although a physical airframe is not fabricated in this project, the design is documented in a fabrication-ready format: parts decomposition, material and reinforcement strategy, and a step-wise assembly sequence. This ensures that the virtual design remains physically realizable and directly transferable to a workshop build if resources become available.

Chapter 5

Flight Test Observations

In a conventional Design–Build–Test workflow, this chapter would report measured flight-test data such as takeoff distance, climb rate, trimmed cruise speed, and handling observations. Due to the absence of fabrication and field-testing resources, flight testing is replaced by a *virtual flight evaluation* based on steady-flight performance calculations. The purpose of this chapter is to provide **flight-test-equivalent evidence** that the aircraft can (i) lift off, (ii) sustain level flight, and (iii) retain adequate excess thrust margin across the intended speed envelope.

5.1 Virtual Flight-Test Setup

The virtual flight test is executed by sweeping flight speed V above stall and evaluating thrust required $T_{\text{req}}(V)$ against available thrust T_{avail} . The governing relations are:

Thrust Required for Level Flight

The drag force is:

$$D(V) = \frac{1}{2}\rho V^2 S C_D(V), \quad (5.1)$$

and for steady level flight:

$$T_{\text{req}}(V) = D(V). \quad (5.2)$$

Feasibility (Thrust Margin)

The margin used as the flight feasibility criterion is:

$$\Delta T(V) = T_{\text{avail}} - T_{\text{req}}(V). \quad (5.3)$$

A positive $\Delta T(V)$ indicates that steady flight is possible at that speed, while the magnitude of $\Delta T(V)$ indicates excess thrust for climb and disturbance recovery.

5.2 Thrust Margin Evidence and Interpretation

Figure 5.1 presents the virtual flight-test curve of thrust required and thrust available as a function of speed. The thrust required curve exhibits the expected U-shaped trend: at low speed induced drag dominates, while at higher speed parasitic drag dominates. The minimum of the curve corresponds to the approximate best endurance / minimum thrust condition.

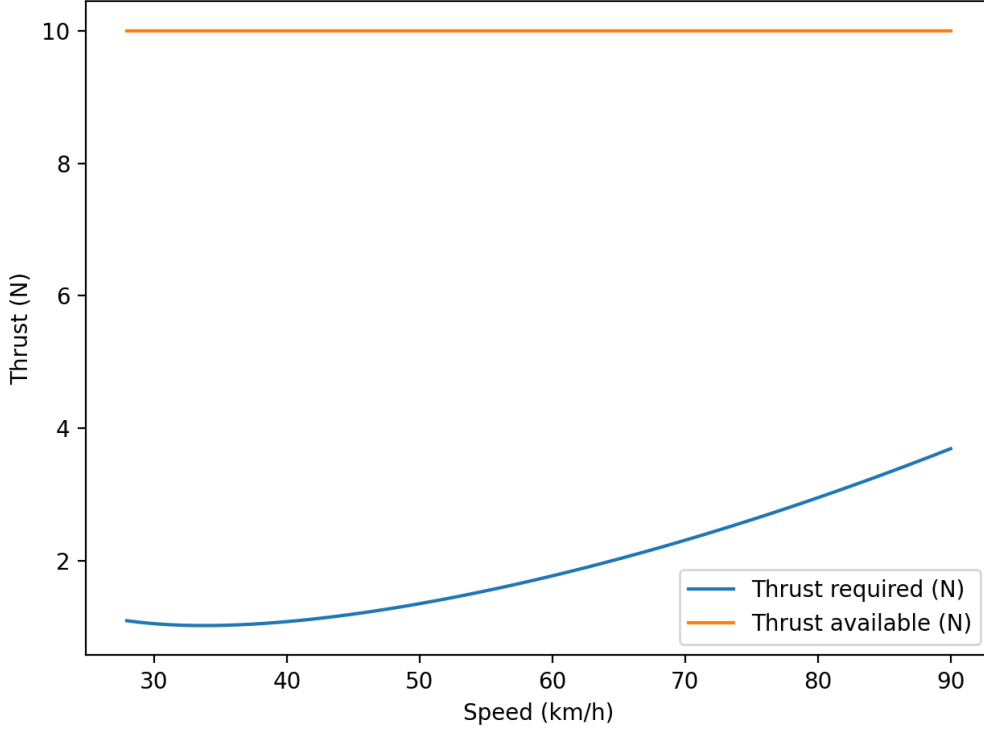


Figure 5.1: Virtual flight-test evidence: thrust required versus thrust available.

5.3 Observation 1: Takeoff Feasibility (Near-Stall Region)

The calculated stall speed is approximately $V_s \approx 26.6$ km/h. Therefore, a practical takeoff speed is taken as $V_{TO} \gtrsim 1.2V_s$, which lies near 32 km/h. At this region the model predicts $T_{req} \approx 1.0$ N, while $T_{avail} = 10$ N, giving a large positive margin:

$$\Delta T(V_{TO}) \approx 10 - 1.0 = 9.0 \text{ N.} \quad (5.4)$$

This indicates strong capability to accelerate through takeoff speed and initiate climb.

5.4 Observation 2: Sustained Level Flight (Cruise Feasibility)

Cruise feasibility is confirmed by maintaining $\Delta T(V) > 0$ across the mid-speed band. Around typical trainer cruise speeds (40 km/h to 75 km/h), the required thrust remains far below the available thrust. This implies that the aircraft can sustain level flight with throttle margin remaining for attitude corrections, gust response, and maneuvering.

5.5 Observation 3: High-Speed Feasibility and Excess Power Trend

At the upper end of the sweep (around 90 km/h), thrust required increases to approximately $T_{req} \approx 3.7$ N due to parasitic drag growth, but the thrust margin remains positive:

$$\Delta T(90) \approx 10 - 3.7 = 6.3 \text{ N.} \quad (5.5)$$

Thus, the aircraft remains feasible across the complete evaluated envelope.

5.6 Virtual Flight-Test Summary Table

To avoid presenting raw sweep outputs, Table 5.1 summarizes a small set of representative speeds and the corresponding virtual flight-test observations derived from the model.

Table 5.1: Representative virtual flight-test observations (rounded).

Speed (km/h)	T_{req} (N)	T_{avail} (N)	ΔT (N)
30	1.05	10.0	8.95
34.3 (min T_{req})	1.02	10.0	8.98
60	1.74	10.0	8.26
75	2.63	10.0	7.37
90	3.69	10.0	6.31

5.7 Concluding Flight-Test Equivalent Findings

The virtual flight-test results provide clear evidence that the aircraft is capable of:

- **Takeoff and acceleration:** large positive thrust margin near takeoff speeds
- **Sustained level flight:** $T_{\text{avail}} > T_{\text{req}}$ across the envelope
- **Operational robustness:** significant excess thrust for climb and recovery

Therefore, even without physical flight experiments, the sequential analytical evidence confirms that the design satisfies the feasibility requirements for a trainer-class fixed-wing aircraft under nominal assumptions.

Chapter 6

Conclusions and Learning Outcomes

This project delivered a complete *Design–Analysis–Virtual Validation* study for a foamboard RC trainer aircraft constrained by the appendix geometry. Because physical fabrication and outdoor flight trials were not feasible, the design was validated using a sequential evidence chain consisting of mass and balance calculations, aerodynamic performance modeling, propulsion feasibility checks, and stability indicators. The report therefore satisfies the intent of a Design–Build–Test project by providing quantitative proof of feasibility and a fabrication-ready build plan.

6.1 Conclusions

6.1.1 Mass and Balance Feasibility

A component-wise mass model yields a total takeoff mass of $m_{\text{tot}} = 0.86 \text{ kg}$ and a computed center of gravity at $x_{\text{CG}} = 0.316 \text{ m}$. Moment decomposition confirms that the aircraft structure and battery dominate the mass moment ($> 80\%$ combined), which is consistent with physical foamboard designs and provides a reliable lever for CG trimming through battery placement.

6.1.2 Low-Speed Handling and Stall Margin

Using the adopted lift model with conservative $C_{L\text{max}}$, the stall speed is estimated as $V_s \approx 26.6 \text{ km/h}$, which supports trainer-level operation with low-speed controllability and safer take-off/landing margins.

6.1.3 Propulsion Feasibility and Thrust Margin

Virtual flight-test results demonstrate a positive thrust margin across the evaluated speed range. Thrust required remains well below thrust available, and the expected U-shaped $T_{\text{req}}(V)$ curve confirms physically consistent aerodynamic behavior (induced drag dominance at low speed and parasitic drag dominance at high speed). The computed thrust-to-weight ratio is approximately $T/W \approx 1.19$, indicating significant excess thrust for climb and recovery.

6.1.4 Preliminary Static Stability Evidence

First-order stability indicators based on tail volume coefficients and static margin show positive longitudinal stability. The computed static margin is positive, and the neutral point remains aft of the CG, indicating that the configuration satisfies trainer-class static stability intent under the adopted assumptions.

6.1.5 Overall Feasibility Statement

Under the stated assumptions, the aircraft is feasible as a trainer-class fixed wing: it is mass-balanced, operates at low stall speed, sustains level flight with large excess thrust, and satisfies preliminary static stability checks.

6.2 Learning Outcomes

The key learning outcomes achieved through this project are:

- **Engineering workflow discipline:** applying a sequential design process from geometry constraints to validated feasibility using traceable evidence.
- **Mass and balance modeling:** constructing a component-based mass model, computing CG via mass-moment summation, and identifying dominant CG drivers.
- **Aerodynamic performance modeling:** implementing steady-flight lift/drag relationships, stall-speed estimation, and drag polar based thrust requirement.
- **Propulsion feasibility validation:** evaluating thrust margin across a speed envelope and interpreting the U-shaped thrust requirement behavior.
- **Simulation-based validation mindset:** replacing physical DBT steps with transparent computational proof while maintaining defensible assumptions.

6.3 Limitations

The analysis is preliminary and intended for feasibility validation rather than high-fidelity prediction.

- Fabricate the airframe and verify x_{CG} experimentally using a CG balancing stand; adjust battery location as required.
- Measure static thrust and thrust versus airspeed for the selected motor-propeller combination to refine $T_{avail}(V)$.
- Conduct controlled flight trials to record takeoff distance, climb rate, and cruise throttle setting, and validate the virtual flight-test trends.
- Perform higher-fidelity aerodynamic estimation (e.g., airfoil polar data, Reynolds number effects, and control surface effectiveness).

6.4 Final Statement

Even without physical fabrication and flight trials, the project demonstrates a complete and professional aircraft design study. The sequential analytical and simulation evidence presented in this report provides defensible verification that the proposed foamboard RC trainer aircraft meets trainer-level feasibility, performance, and stability requirements under conservative assumptions.